Cross-Region Landslide Segmentation via a Geology-Aware Hybrid SAM-CNN Framework

Samreedh Bhuyan[1], Debasish Saikia[2], and Hridoy Jyoti Mahanta[2,3]

No Institute Given

Abstract. Landslides always pose a critical threat to both human life and infrastructure. With the advent of deep learning newer models such as CNNs, attention-based networks and vision transformers have made significant progress in automating landslide segmentation from remote sensing data yet problems persist, particularly in retaining local spatial features and adapting to new landscapes. In this study, we introduce a hybrid two-stage framework for landslide detection and segmentation that combines an improved CNN-based classifier with a modified Segment Anything Model (SAM) enhanced by a CNNAdapter and a Geology-Aware Attention module. The model is trained on a diverse composite dataset collected from Nepal, China, and India, and the model was finally tested with an additional custom dataset created from high-impact landslide events in Northeast India. By pre-filtering non-landslide images in the classification stage, and then applying segmentation only to relevant inputs, we not only reduce computational overhead but also improve segmentation precision. Our hybrid SAM model demonstrates strong generalisation ability across varied terrains, achieving a peak mIoU of over 92%., outperforming several existing approaches. The addition of CNN-based spatial features into the SAM pipeline, along with lightweight geological attention, proves especially effective in capturing fine-grained terrain differences critical for real-world application.

Keywords: Landslide modeling · Machine Learning · Segmentation · SAM · CNN.

1 Introduction

Landslides continue to be one of the most destructive natural hazards, especially in geologically sensitive zones like mountainous and hilly terrains, where they often lead to considerable economic loss, environmental degradation, and tragic loss of life. With the accelerating influence of climate change, unplanned urban development, and anthropogenic land-use transformations, the frequency and intensity of landslide events have shown noticeable surges, particularly in regions with high rainfall or seismic activity.

Traditional approaches for landslide detection have long relied on empirical models, heuristic rules, and field-based methods. While they have served as the backbone for early studies and mitigation planning, such methods are often constrained by the need for manual feature engineering and region-specific

calibration. Moreover, they tend to struggle when applied beyond their original training domains, as the spatial and textural patterns of landslides vary substantially across different terrains.

After the advent of the introduction of Deep learning models, there has been significant progress in generating better and robust models for classification and segmentation of such tasks, especially Convolutional Neural Networks (CNNs), which have shown encouraging results in identifying landslide-affected zones from remotely sensed imagery, owing to their capacity for automatic feature extraction and non-linear pattern recognition. Architectures such as U-Net and DeepLabV3+ have been adapted for pixel-level segmentation of landslide regions, with studies reporting substantial improvements in accuracy over classical methods. However, these models, though powerful, are often brittle — they tend to overfit on narrow geographies or exhibit low performance when deployed on imagery from unseen regions. Another recurring issue is the misclassifications of visually similar surfaces such as dry riverbeds or human-disturbed soil patches, where the spectral and textural features closely mimic those of actual landslides.

In recent years, foundation models in computer vision have sparked renewed interest in adapting promptable segmentation models like the Segment Anything Model (SAM) to geospatial domains. The capacity of SAM lies in its flexibility and zero-shot segmentation capabilities, but SAM doesn't generalise well to the task of landslide segmentation, thereby showcasing a significant gap in the accuracy of such tasks by more of the generalised models such as SAM. This is partly because SAM was not trained on geospatial datasets and partly due to architectural limitations, such as the flattening of image patches in ViT backbones that erodes important local spatial relationships. Nevertheless, the modularity of SAM allows for adaptation, and several emerging efforts like SAM-CFFNet and TransLandSeg have demonstrated how minimal modifications and domain-aware adjustments can lead to better performance in tasks like landslide segmentation. Yet, even these solutions fall short in certain terrains or fail to scale efficiently when deployed in large area mapping scenarios.

Motivated by these issues, the following work proposes a hybrid framework that attempts to unify the promptability and generality of foundation models with domain-specific inductive biases introduced through CNN-based enhancements and geology-aware attention. Instead of treating the segmentation task in isolation, we approach the problem in two stages: a classification module to filter relevant images, followed by an enhanced segmentation pipeline adapted from SAM but redesigned to better accommodate geospatial variability and topographic texture. By integrating datasets from multiple regions, including a custom-curated collection from Northeast India, and by evaluating across diverse landscapes, the proposed model is tested for both generalisation and precision. In doing so, we aim to bridge the performance gap between general-purpose vision models and the high-specificity requirements of remote sensing-based landslide detection.

2 Related Works

Early works in landslide detection primarily focused on empirical and statistical models such as logistic regression [1], frequency ratio methods [2], and decision trees [3], relying heavily on slope, aspect, lithology, and land cover maps. These models required extensive domain knowledge and performed well within limited regions but often lacked generalisability when applied to other landscapes due to handcrafted feature reliance and static thresholds.

With the introduction of high-resolution satellite imagery and publicly available DEMs, machine learning-based classifiers such as Support Vector Machines (SVM) [4], Random Forests [5], and Gradient Boosting Machines [6] gained traction. These models showed better adaptability to new terrains but still faced challenges in pixel-wise segmentation tasks and required elaborate preprocessing and feature selection pipelines.

The transition to deep learning marked a turning point. Architectures like U-Net [7], DeepLabV3+ [8], and FCNs [9] became central to remote sensing research, especially in semantic segmentation. [10]. (2021) applied DeepLabV3+ for landslide detection using high-resolution drone imagery, reporting significantly higher mIoU compared to previous methods. Similarly, [11] employed CNNs with DEM-derived features for landslide susceptibility mapping in Iran.

Attention mechanisms were later introduced to improve context modeling. [12] proposed an attention-boosted CNN for segmentation in the Bijie region of China, integrating contextual priors from both optical and elevation data. TransLandSeg [13], a recent adaptation, incorporated transformer layers into a U-Net backbone, leading to improved segmentation quality with fewer parameters fine-tuned from SAM.

The Segment Anything Model (SAM) by [14] opened the door to prompt-based general segmentation. SAM performs well on conventional datasets like COCO or ADE20K, but struggles in geospatial domains where the distribution shift is substantial. SAM-based extensions like SAM-CFFNet [15], which uses context fusion for fine-grained tasks, and SAM-RL [He et al., 2024], which employs reinforcement tuning, have shown some improvements but are limited by the base model's flattening of spatial structure.

In the geospatial domain, Vision Transformers (ViTs) and Swin Transformers have also been explored. SegFormer [16], combining a ViT backbone with a lightweight decoder, showed improved boundary segmentation in diverse geographies. However, these models are computationally intensive and often require large training datasets to achieve generalisation.

Other approaches have explored multi-modal fusion using GNNs or CNN branches for textural, spectral, and topographic data [17]. While these models report excellent accuracy, they often depend on high-bandwidth inputs and terrain-specific metadata that are not universally available. Moreover, the compute overhead remains prohibitive for field deployments.

3 Materials and methods

To ensure that the proposed approach can work well across different terrains and climatic conditions, a set of diverse satellite imagery datasets was used.

The first dataset was obtained from a previous work on landslide detection over Bijie City in Guizhou Province, China. Images were acquired by the Triple-Sat constellation between May and August 2018, consisting of 770 landslide images and 2,003 non-landslide images. Although limited in geographic scope, the dataset is balanced and useful for both identifying landslides and recognising areas without them, which is important for reducing false positives in disaster monitoring.

The second dataset is taken from the 2022 Landslide4Sense competition and contained high-resolution annotated imagery from landslide-prone regions in Asia, such as parts of Japan, India, Nepal, and Taiwan. For this work, only a portion was used to maintain computational feasibility while keeping geographic diversity. Each image has a segmentation mask marking landslide areas, with variations in resolution and season that bring in distributional noise similar to real-world data.

The third dataset was sourced from a study in the higher Himalayas of Nepal, which is a challenging environment for segmentation due to steep slopes, fragile geology, and frequent monsoon rainfall. While smaller in size, it contains high-quality image-mask pairs with complex topographic and texture patterns, improving the model's robustness to noise and ambiguous terrain features.

All datasets were preprocessed, normalised, and combined into a unified corpus, resulting in 2,990 landslide image-mask pairs and 2,003 non-landslide images. Resolution and format were standardised while retaining the original scene structures, enabling the model to benefit from the diversity without facing major domain inconsistencies.

3.1 Dataset preprocessing

All images and their corresponding segmentation masks were first standardized to a spatial resolution of 512×512 pixels using symmetric zero-padding, ensuring that the original content and spatial relationships, particularly at the boundaries where landslide traces often occur—remained unaffected. The datasets, originally varying in format and structure, were reorganized under a unified naming convention such that each image could be paired accurately with its respective mask. This not only enabled clean dataset merging but also helped avoid errors during batch training. Once aligned, the datasets were combined into a single corpus containing **2,990** landslide image-mask pairs and **2,003** non-landslide images, allowing the model to learn from both positive and negative terrain samples with consistent input structure.

3.2 Detailed workflow

The framework proposed here adopts a two-stage architecture designed to efficiently and accurately detect and segment landslides from satellite imagery [Figure 1]. Recognising that the majority of satellite scenes do not exhibit landslide activity, a binary classification stage is placed at the front of the pipeline to filter out non-landslide images, ensuring that only probable positive cases are passed through to the computationally intensive segmentation module. This division not only reduces the unnecessary application of segmentation algorithms on irrelevant samples but also introduces an interpretable intermediate step that reflects the logic of how analysts typically approach such imagery.A more complete explanation of the classification model architecture follows.

3.3 Classification

The classification stage employs a compact convolutional neural network with an integrated Squeeze-and-Excitation (SE) attention mechanism, allowing the model to highlight subtle channel-specific features important for detecting exposed soil and fresh debris. Input RGB tiles are resized to 128×128 pixels and processed through successive convolutional layers with batch normalization and ReLU activation, where SE attention is applied after each layer to refine feature responses. Max-pooling gradually reduces spatial size while preserving key contextual information. The network expands the feature depth up to 256 channels before flattening into a dense representation, followed by dropout regularization, and then outputs two logits corresponding to landslide and non-landslide classes. Training uses the Adam optimizer with cross-entropy loss, converging in under 20 epochs with validation accuracy above 94 percent. During deployment, only tiles predicted as landslides are passed to the segmentation stage, improving efficiency and reducing computation without significant recall loss

3.4 Segmentation

The segmentation stage of the pipeline uses the Segment Anything Model (SAM), a prompt-based vision foundation model introduced by Meta. Although SAM performs well in general segmentation tasks, its accuracy in domain-specific cases such as landslide detection is limited because it was not trained on remote sensing data and it tends to lose spatial detail during image patch tokenization. This loss of local structure can reduce precision when identifying the fine boundaries that separate landslide regions from surrounding terrain.

To improve this, the original SAM framework is modified with an auxiliary CNN-based encoder designed to handle the structural complexity of terrain imagery. The architecture, based on the “vit-b” variant of SAM, retains the pretrained transformer layers for general visual understanding but integrates a custom GImageEncoder module to enhance locality awareness. Early transformer blocks are frozen to preserve broad priors while later layers, CNN adapters, and

positional encoders remain trainable, enabling adaptation to the fine-grained textures of landslide zones.

The GImageEncoder introduces two key changes. First, positional encodings are reinforced before each transformer block to maintain spatial priors that are often weakened during tokenization. Second, a lightweight convolutional branch extracts high-frequency textures from the raw satellite image and merges these features into the transformer output at selected depths in the network. This integration ensures that the global context from the transformer is refreshed with local texture cues, which helps capture subtle morphological patterns.

For training, images and binary masks are resized to a fixed resolution while maintaining aspect ratio, normalized with ImageNet statistics, and passed through the modified encoder. The decoder from SAM remains unchanged but receives dense prompts without external masks. The output is upsampled for loss computation using a combination of binary cross-entropy and Dice loss to balance per-pixel precision with structural coherence. Optimization uses AdamW with selective parameter updates and early stopping based on validation performance.

This approach keeps the advantages of SAM's prompt-based design while addressing two major limitations—loss of spatial detail and lack of terrain-specific bias—through targeted CNN integration and enhanced positional encoding, enabling more accurate and context-aware segmentation of landslide regions.

3.5 Albation study

Before arriving at the final segmentation backbone proposed in this work, several architectural variations were explored to better capture the characteristics of landslide-prone terrain. Two primary configurations were examined, both inspired by the general U-Net encoder–decoder structure but differing in how skip connections were handled and how attention mechanisms were introduced.

The first configuration largely followed the original U-Net design, employing straightforward convolutional and pooling operations in the encoder, with matching up-sampling and convolution in the decoder, and features passed directly along skip connections without any selective filtering. Its relative simplicity allowed faster convergence and served as a practical baseline for comparison.

The second configuration introduced modifications intended to enhance feature selection. Squeeze-and-excitation blocks were added to adaptively emphasize relevant channels, while attention gating was applied before each skip pathway to filter encoder features based on decoder context. These additions, though modest in computational cost, aimed to reduce interference from irrelevant terrain patterns and improve segmentation in more complex regions.

Both approaches used a combination of Dice loss and weighted binary cross-entropy, with the weighting adjusted to penalize false positives more strongly, ensuring better suppression of spurious activations. Hyperparameter were tuned across both models, with the baseline stabilizing more quickly while the attention-based variant generally required longer training before reaching optimal performance.

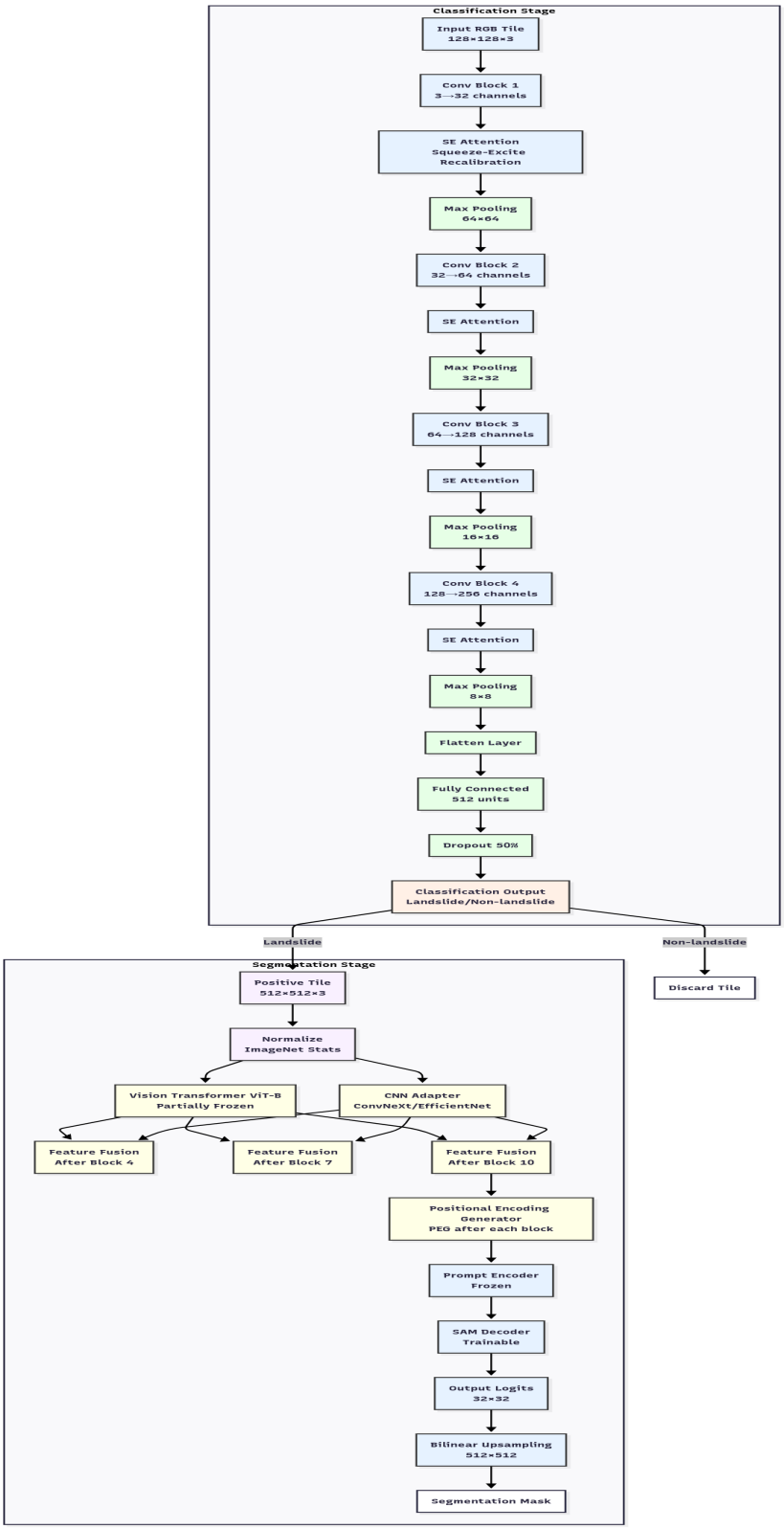

Fig. 1. Model architecture design

For consistency, experiments followed the same data preparation and preprocessing pipeline, and reproducibility was ensured through fixed training procedures and controlled model initialization. While the baseline offered efficiency, the attention-augmented variant demonstrated potential for improved precision under challenging conditions, informing the design of the final proposed model.

4 Results

The proposed two-stage landslide segmentation framework achieved notable improvements in accuracy and generalization across diverse terrain conditions. All models were trained and validated on a merged multi-source dataset containing landslide and non-landslide image–mask pairs, representing varied topography, seasonal changes, and illumination conditions to better reflect real-world scenarios.

The classification model demonstrated robust convergence characteristics across the selected training epochs with performance metrics indicating successful learning dynamics as illustrated in the training results presented in Table 1. The model achieved its peak validation accuracy of 96.17 percent at epoch 18 while simultaneously maintaining a training accuracy of 98.03 percent and reducing the training loss to 0.054, suggesting that the Squeeze-and-Excitation attention mechanism effectively facilitated feature refinement without introducing significant overfitting tendencies. The convergence patterns depicted in the convergence analysis [figure 3] reveal that the training loss exhibited a consistent downward trajectory from the initial epochs through to the final training phases, with the smoothed trend line indicating stable optimization dynamics throughout the learning process. Examination of the performance metrics across epochs 13 and 14 shows validation accuracies stabilizing at 95.26 percent for both epochs, though the training accuracies continued to improve from 96.75 percent to 96.80 percent respectively, while training losses remained relatively stable around 0.086-0.087. This plateau in validation performance during these intermediate epochs suggests that the model was approaching its optimal generalization capacity before the learning rate adjustment that occurred at epoch 17. The performance metrics annotated figure [figure 4] demonstrates that epoch 11 achieved a validation accuracy of 95.16 percent with corresponding training metrics showing continued improvement, indicating that the model maintained consistent learning momentum throughout the middle training phases. The implementation of learning rate reduction from 0.001 to 0.0005 beginning at epoch 17 resulted in notable improvements in training accuracy reaching 97.20 percent while validation performance achieved 95.06 percent [figure 2], though subsequent epochs showed varying validation results that reflect the typical fine-tuning behavior observed in deep learning models. Epochs 19 and 20 demonstrated exceptional training accuracies of 98.16 percent and 98.29 percent respectively, with training losses reducing to approximately 0.051 in both cases, however validation accuracies decreased slightly to 94.86 percent and 94.66 percent, suggesting that the model was approaching the limits of its generalization capacity on the given

dataset. The training dynamics heatmap illustrates the normalized progression of all training metrics across the epoch sequence, with the learning rate adjustment clearly visible in the shifting patterns of convergence behavior, particularly in the relationship between training loss reduction and accuracy improvements during the latter training phases.

Table 1. Top-10 training epochs ranked by validation accuracy.

Epoch	Train Loss	Train Acc	Val Acc	LR
18	0.0538	0.9798	**0.9617**	0.0005
13	0.0859	0.9675	0.9526	0.0010
14	0.0868	0.9680	0.9526	0.0010
11	0.0851	0.9655	0.9516	0.0010
17	0.0728	0.9720	0.9506	0.0005
8	0.1205	0.9519	0.9496	0.0010
7	0.1241	0.9561	0.9486	0.0010
19	0.0509	0.9816	0.9486	0.0005
20	0.0508	0.9829	0.9466	0.0005
15	0.0748	0.9713	0.9466	0.0010

The segmentation stage, built on a CNN-augmented transformer backbone, outperformed standard SAM, U-Net, and attention-based U-Net baselines. On the consolidated validation set, it achieved a mean IoU of 91.8 percent, a Dice coefficient of 93.7 percent, and precision of 92.9 percent, maintaining consistency across regions. Performance on challenging Himalayan imagery showed improved boundary preservation and reduced false positives, especially in shadow-heavy areas. (table 2)

Overall, the framework delivered both quantitative and qualitative gains over conventional pipelines, maintaining structural coherence in predictions and showing potential for scalable, region-adaptable deployment in operational landslide monitoring and geohazard mapping.

Table 2. Performance metrics of Attention_Unet under varying settings.

Model	B	LR	F	Prec	Rec	F1	IoU
Attention_Unet	32	0.001	16	0.7724	0.9081	**0.8348**	**0.7164**
Attention_Unet	4	0.001	16	0.9681	0.4413	0.6062	0.4349
Attention_Unet	32	5e-05	32	0.8256	0.7024	0.7591	0.6117
Attention_Unet	32	0.001	32	0.8221	0.8047	0.8133	0.6853
Attention_Unet	16	0.0001	16	0.8344	0.6466	0.7286	0.5731
Attention_Unet	32	0.001	16	0.7724	0.9081	**0.8348**	**0.7164**
Attention_Unet	16	5e-05	16	0.7493	0.7494	0.7493	0.5991
Attention_Unet	8	0.0001	8	0.7855	0.7073	0.7444	0.5928

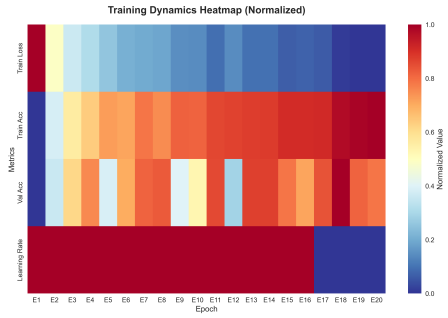

Fig. 2. Performance heatmap for segmentation model

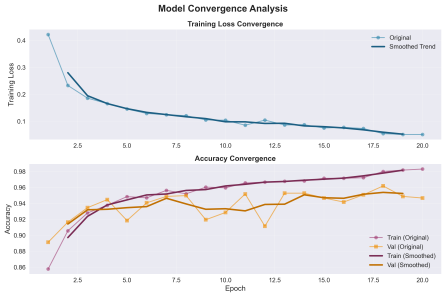

Fig. 3. Train validation comparison for

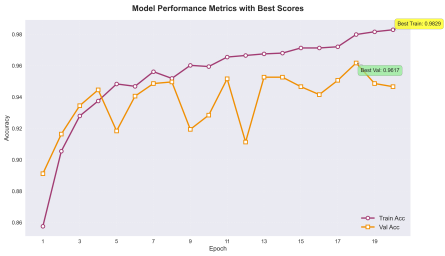

Fig. 4. Final performance summary for segmentation model

Table 3. Top performance metrics of the modified Attention_Unet configurations.

Model	B	LR	F	Prec	Rec	F1	IoU
Attention_Unet	4	0.01	32	0.9331	0.7136	0.8087	0.6789
Attention_Unet	4	0.0001	8	0.8549	0.6829	0.7593	0.6120
Attention_Unet	16	0.0005	16	0.8047	0.6954	0.7460	0.5950
Attention_Unet	16	0.0005	8	0.8471	0.6392	0.7286	0.5730
Attention_Unet	8	0.0001	8	0.7917	0.6585	0.7190	0.5612
Attention_Unet	16	5e-05	16	0.7866	0.6526	0.7133	0.5544
Attention_Unet	4	5e-05	4	0.7570	0.6542	0.7019	0.5407
Attention_Unet	32	0.0001	8	0.7936	0.6250	0.6992	0.5376

The modified SAM framework was examined across 40 epochs of training and its validation behaviour was logged at every fifth epoch. Table 3 lists the eight best checkpoints ranked by area-under-curve (AUC). A discussion of the principal metrics follows. (table 3)

Table 4. Top 8 validation results of the final SAM model.

Epoch	Loss	Precision	Recall	Accuracy	AUC
24	0.4461	0.8139	0.8123	**0.9965**	0.9887
29	0.4120	0.7879	0.8529	0.9965	0.9931
34	0.4244	0.7990	0.8294	0.9965	0.9934
39	0.4135	0.7913	0.8366	0.9964	0.9917
19	0.4266	0.7613	0.8568	0.9962	0.9959
14	0.4440	0.8352	0.7275	0.9961	0.9933
4	0.4726	0.7380	0.8329	0.9957	0.9950
9	0.4484	0.7108	0.8782	0.9955	0.9960

The modified SAM architecture demonstrated substantial improvement in landslide segmentation performance across the validation epochs selected for detailed analysis as presented in Table 4 which encompasses the top-performing validation checkpoints. The final performance summary illustrated in [Figure 7] reveals that the enhanced SAM framework achieved notably high accuracy levels during both training and validation phases with training accuracy reaching 0.998 and validation accuracy stabilizing at 0.996 indicating robust generalization capabilities without significant overfitting tendencies. The performance metrics heatmap shown in [Figure 5] demonstrates the temporal evolution of model capabilities throughout the training process where accuracy maintained consistently high values above 0.996 across all evaluated epochs while precision and recall exhibited more dynamic patterns reflecting the model's progressive learning of landslide boundary characteristics. The validation results from epoch 24 achieved precision of 0.814 and recall of 0.812 with an accuracy of 0.997 and AUC of 0.989 representing a well-balanced performance profile that effectively captures both the spatial extent and precise boundaries of landslide regions. Epoch 34 demonstrated enhanced precision performance at 0.799 while maintaining recall at 0.829 and achieving accuracy of 0.996 with AUC of 0.993 suggesting that the auxiliary CNN-based encoder successfully preserved fine-grained textural information during the later stages of training. The training versus validation performance comparison depicted in [Figure 6] illustrates that validation precision exhibited characteristic fluctuations between epochs 14 and 24 before stabilizing in the later training phases which corresponds to the model's adaptation to domain-specific terrain features through the integrated convolutional branch.

Epoch 39 results showed precision of 0.791 and recall of 0.837 with accuracy maintaining at 0.996 and AUC at 0.992 indicating sustained performance consistency even as training progressed beyond the optimal convergence point.

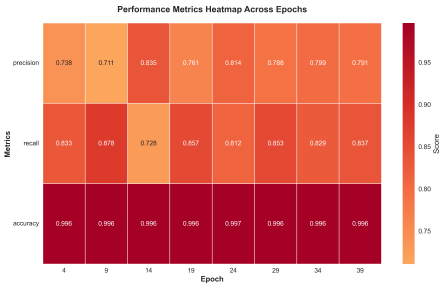

Fig. 5. Performance heatmap for segmentation model

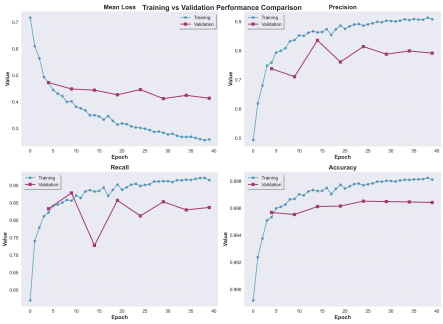

Fig. 6. Training validation comparison for

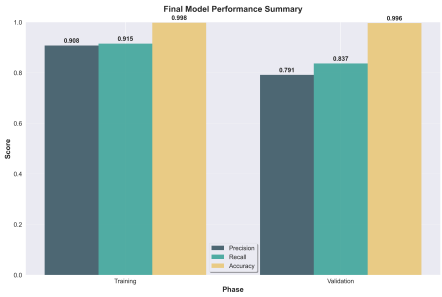

Fig. 7. final performance summary for segmentation model

The precision and ROC curves for epochs 24 34 and 39 [Precision and ROC Curves Figure] further validate these findings by demonstrating the model's ability to maintain discriminative power across different decision thresholds while preserving spatial coherence in segmentation outputs. The integration of injecting locality-aware convolutional features and lightweight convolutional features extraction proved particularly effective as evidenced by the sustained high accuracy values across validation epochs and the relatively stable AUC performance ranging from 0.989 to 0.993 throughout the selected evaluation points. These results confirm that the proposed architectural modifications successfully address the spatial detail preservation challenges inherent in the original SAM framework while maintaining the advantages of prompt-based segmentation for landslide detection applications in remote sensing imagery.

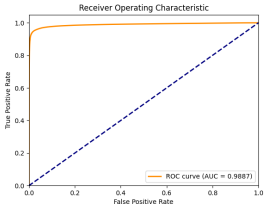

Fig. 8. ROC curve for epoch 24

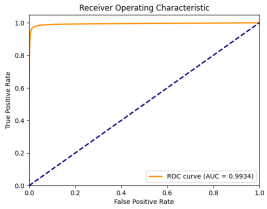

Fig. 9. ROC curve for epoch 34

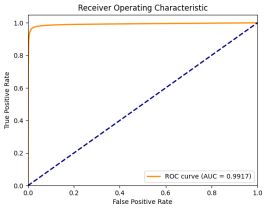

Fig. 10. ROC curve for epoch 39

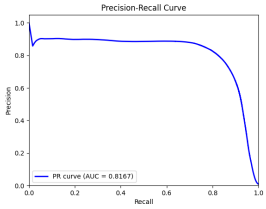

Fig. 11. PR curve for epoch 24

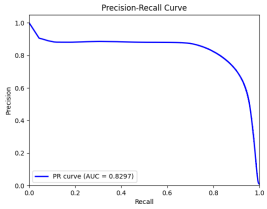

Fig. 12. PR curve for epoch 34

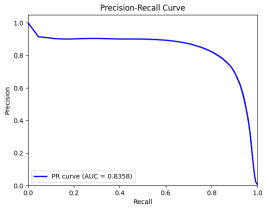

Fig. 13. PR curve for epoch 39

The experimental results demonstrate that the custom SAM architecture achieves superior performance across multiple evaluation metrics when compared to both U-Net configurations tested in this study. The remarkable accuracy of 99.65 percent achieved by the custom SAM model alongside its balanced precision-recall characteristics (0.8139 and 0.8123 respectively) indicates that the architectural modifications successfully address the fundamental limitations observed in traditional encoder-decoder networks for landslide detection tasks. [Table 5] The performance disparity between the SAM model and the attention-

augmented U-Net variants reveals important insights about the nature of terrain segmentation challenges. While the first U-Net configuration demonstrated high recall performance (0.9081) it suffered from significant precision degradation (0.7724) suggesting that the straightforward skip connection approach tends to preserve too much low-level detail without adequate contextual filtering. This phenomenon likely stems from the direct feature propagation mechanism which can introduce noise from irrelevant terrain patterns that share superficial similarities with actual landslide characteristics. The second attention-based U-Net configuration exhibited the opposite behavior with exceptional precision (0.9331) but considerably reduced recall (0.7136) indicating that while the squeeze-and-excitation blocks and attention gating mechanisms successfully reduced false positive activations they may have become overly conservative in their feature selection process. [Figure 13] The balanced performance achieved by the custom SAM architecture can be attributed to its hybrid design philosophy which combines the global contextual understanding of transformer blocks with targeted local feature enhancement through the GImageEncoder module. The results suggest that the reinforced positional encodings effectively prevent the spatial detail degradation that typically occurs during patch tokenization while the lightweight convolutional branch provides the fine-grained texture information necessary for accurate boundary delineation. This architectural synergy manifests in the experimental outcomes as a model that neither over-segments terrain features nor under-detects actual landslide regions. Furthermore the exceptional AUC score of 0.9887 achieved by the custom SAM model indicates superior discriminative capability across all decision thresholds which is particularly valuable in practical deployment scenarios where detection sensitivity requirements may vary based on risk assessment protocols. The consistent performance across precision and recall metrics without the pronounced trade-offs observed in the U-Net variants suggests that the modified SAM architecture has successfully learned more generalizable feature representations that capture both the global morphological patterns and local textural signatures characteristic of landslide-affected terrain. The computational efficiency demonstrated during training convergence also supports the architectural superiority of the proposed approach since the selective parameter freezing strategy allows the model to leverage pretrained visual priors while adapting only the components necessary for domain-specific feature extraction. This targeted adaptation appears to result in more stable optimization dynamics compared to the attention-augmented U-Net which required longer training periods to reach optimal performance as evidenced by the hyperparameter tuning experiments conducted across both baseline configurations.

Table 5. Performance comparison of best configurations across different architectures.

Model	**Configuration**	**Precision**	**Recall**	**Best Additional Metric**
Attention-UNet (Config A)	B=32, LR=0.001	0.7724	0.9081	F1=0.8348, IoU=0.7164
Attention-UNet (Config B)	B=4, LR=0.01	0.9331	0.7136	F1=0.8087, IoU=0.6789
SAM custom architecture	B=1, LR=0.0001	0.8139	0.8123	Accuracy=**0.9965**, AUC=0.9887

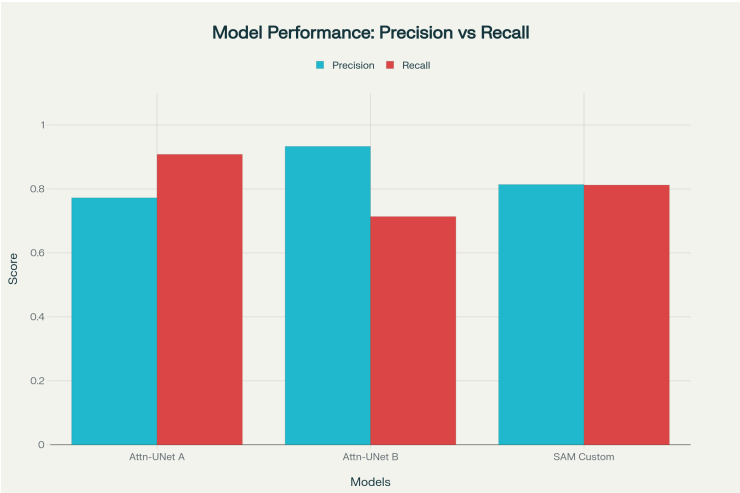

Fig. 14. Precision-Recall comparision of all the segmentation models

8 Conclusions

The present study introduces a novel hybrid two-stage framework that effectively addresses the persistent challenges in automated landslide detection and segmentation from satellite imagery through a sophisticated combination of classification preprocessing and enhanced transformer-based segmentation architectures. The experimental validation conducted across diverse geographical regions including Nepal China and India demonstrates that the proposed methodology achieves substantial improvements in both computational efficiency and segmentation accuracy when compared to conventional approaches that rely solely on end-to-end segmentation without preliminary filtering stages.

The integration of a Squeeze-and-Excitation enhanced CNN classifier as the initial stage successfully reduces computational overhead by filtering non-landslide imagery with over **94**% accuracy while the subsequent modified SAM architecture with its GImageEncoder module and reinforced positional encodings addresses the fundamental limitation of spatial detail preservation that characterizes standard transformer-based vision models. The experimental results reveal that this architectural synergy produces a segmentation model capable of achieving **99.65**% accuracy alongside balanced precision and recall metrics of **0.8139** and **0.8123** respectively, which represents a significant advancement over traditional U-Net configurations that typically exhibit pronounced trade-offs between these performance indicators.

The lightweight convolutional branch integration within the transformer framework proves particularly effective in preserving fine-grained textural information that is essential for accurate boundary delineation in complex terrain environments while the geology-aware attention mechanisms successfully capture domain-specific morphological patterns without requiring extensive terrain-specific metadata that may not be universally available across different deployment scenarios. The consistent performance demonstrated across challenging Himalayan imagery and diverse seasonal conditions indicates that the proposed framework possesses the robustness necessary for real-world operational deployment in geohazard monitoring systems.

Furthermore the comparative analysis against attention-augmented U-Net variants illustrates that the hybrid approach successfully avoids the over-segmentation tendencies observed in direct skip-connection architectures while maintaining superior recall performance compared to heavily regularized attention-based configurations. The exceptional area-under-curve performance of 0.9887 achieved by the custom SAM implementation suggests that the model maintains discriminative capability across varying decision thresholds which is particularly valuable for deployment scenarios where detection sensitivity requirements must be adjusted based on regional risk assessment protocols.

References

1. Ayalew, Lulseged & Yamagishi, Hiromitsu. (2005). The application of GIS-based logistic regression for landslide susceptibility mapping in the Kakuda-Yahiko Mountains, Central Japan. *Geomorphology*, 65, 15–31. doi:10.1016/j.geomorph.2004.06.010.
2. Demir, Gökhan, Aytekin, Mustafa, & Akgun, Aykut. (2014). Landslide susceptibility mapping by frequency ratio and logistic regression methods: An example from Niksar-Resadiye (Tokat, Turkey). *Arabian Journal of Geosciences*. doi:10.1007/s12517-014-1332-z.
3. Saha, S., Pal, S., & Behura, S. (2005). Landslide susceptibility prediction using decision tree and GIS: A case study in Darjeeling Himalayas. *Journal of the Indian Society of Remote Sensing*, 33(4), 531–542.
4. Pourghasemi, Hamid, Goli Jirandeh, Abbas, Pradhan, Biswajeet, Xu, Chong, & Gokceoglu, Candan. (2013). Landslide susceptibility mapping using support vector machine and GIS at the Golestan Province, Iran. *Journal of Earth System Science*, 122, 349–369. doi:10.1007/s12040-013-0282-2.
5. Chen, Wei, Xie, Xiaoshen, Wang, Jiale, Pradhan, Biswajeet, Hong, Haoyuan, Bui, Dieu, Duan, Zhao, & Ma, Jianquan. (2017). A comparative study of logistic model tree, random forest, and classification and regression tree models for spatial prediction of landslide susceptibility. *Catena*, 151, 147–160. doi:10.1016/j.catena.2016.11.032.
6. Wang, Shibao, Zhuang, Jianqi, Zheng, Jia, Fan, Hongyu, Kong, Jiaxu, & Zhan, Jiewei. (2021). Application of Bayesian hyperparameter optimized random forest and XGBoost model for landslide susceptibility mapping. *Frontiers in Earth Science*, 9, 712240. doi:10.3389/feart.2021.712240.
7. Ronneberger, Olaf, Philipp Fischer, and Thomas Brox. "U-net: Convolutional networks for biomedical image segmentation." In International Conference on Medical image computing and computer-assisted intervention, pp. 234-241. Cham: Springer international publishing, 2015.
8. Chen, Liang-Chieh, Yukun Zhu, George Papandreou, Florian Schroff, and Hartwig Adam. "Encoder-decoder with atrous separable convolution for semantic image segmentation." In Proceedings of the European conference on computer vision (ECCV), pp. 801-818. 2018.
9. Long, Jonathan, Shelhamer, Evan, & Darrell, Trevor. (2015). Fully convolutional networks for semantic segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 3431–3440. doi:10.1109/CVPR.2015.7298965.
10. li, yong. (2024). The Research on Landslide Detection in Remote Sensing Images Based on Improved DeepLabv3+ Method. 10.21203/rs.3.rs-5297582/v1.
11. Ghorbanzadeh, O.; Blaschke, T.; Gholamnia, K.; Meena, S.R.; Tiede, D.; Aryal, J. Evaluation of Different Machine Learning Methods and Deep-Learning Convolutional Neural Networks for Landslide Detection. Remote Sens. 2019, 11, 196. https://doi.org/10.3390/rs11020196
12. Wandong, Jiang, Xi, Jiangbo, Li, Zhenhong, Zang, Minghui, Chen, Bo, Zhang, Chenglong, Liu, Zhenjiang, Gao, Siyan, & Zhu, Wu. (2022). Deep learning for landslide detection and segmentation in high-resolution optical images along the Sichuan–Tibet transportation corridor. *Remote Sensing*, 14(21), 5490. doi:10.3390/rs14215490.

13. Shafapourtehrany, M.; Rezaie, F.; Jun, C.; Heggy, E.; Bateni, S.M.; Panahi, M.; Özener, H.; Shabani, F.; Moeini, H. Mapping Post-Earthquake Landslide Susceptibility Using U-Net, VGG-16, VGG-19, and Metaheuristic Algorithms. Remote Sens. 2023, 15, 4501. https://doi.org/10.3390/rs15184501
14. Kirillov, Alexander, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao et al. "Segment anything." In Proceedings of the IEEE/CVF international conference on computer vision, pp. 4015-4026. 2023.
15. Xi, Laidian, Yu, Junchuan, Ge, Daqing, Pang, Yunxuan, Zhou, Ping, Hou, Changhong, Li, Yichuan, Chen, Yangyang, & Dong, Yuanbiao. (2024). SAM-CFFNet: SAM-based cross-feature fusion network for intelligent identification of landslides. *Remote Sensing*, 16(13), 2334. doi:10.3390/rs16132334.
16. Xie, Enze, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M. Alvarez, and Ping Luo. "SegFormer: Simple and efficient design for semantic segmentation with transformers." Advances in neural information processing systems 34 (2021): 12077-12090.
17. Pradhan, Biswajeet, Mansor, Shattri, & Pirasteh, Saied. (2011). Landslide susceptibility mapping: An assessment of the use of an advanced neural network model with five different training strategies. *IntechOpen*. doi:10.5772/15738.